

Adjustment variables in linear regression and logistic regression

CRISP R Mini-Course

2026

Day 7

Review from last time

We wish to ^{predictor} determine the association between **disease severity score** and **treatment status (binary: treated/untreated)**. We run the following code in R, and obtain the output shown.

Explain what this code does and how to interpret the results.

```
# df is a dataframe with age, tx (treated/untreated)
mod <- lm(disease_severity ~ tx, data = df)

summary(mod)
confint(mod)
```

linear regression
formula
dataframe

Model

*Disease Severity =
 $\beta_0 + \beta_1 \text{ treatment}$*

outcome
↓

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.72527	0.87731	-0.827	0.40881
txTreated	1.43562	0.44416	3.232	0.00131 **

	2.5 %	97.5 %
(Intercept)	-2.448965779	0.99843098
txTreated	0.562955300	2.30828457

The difference in severity score
b/w treated vs untreated is 1.43 (95% CI: 0.56 - 2.31)

Mean disease severity for
untreated is -0.72

Announcements

- Friday's tutorial updated with linear regression with categorical variables (including how to reorder factor variables)
- Homework 3 (final exam) due on Thursday, July 23
- This week's schedule:
 - **Today:** Logistic regression
 - **Wednesday:** Plotting using `ggplot2`, using AI tools, and review
- Will post additional tutorials on website on Wednesday

Today's agenda

↙ numeric / continuous
outcomes

- Adding adjustment variables in linear regression

- univariate: one variable

- predictors: binary / categorical or continuous

- adjust?

- Logistic regression – conceptual tutorial

- Running and interpreting logistic regression in R

Linear regression w/ adjustment variable (1)

- We are trying to determine if a ^{blood pressure} ~~disease severity score~~ is associated with **treatment status** adjusting for **age**. We obtain data from an observational study with all variables.  why adjust?
- We answer this question using linear regression:
 - ① age might be a confounder
 - ② randomized trial
"increased precision"
narrower confidence interval
- We do this in R and obtain the following output:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	91.86909	1.13694	80.804	<2e-16	***
treatmentTreated	-0.58538	0.77626	-0.754	0.451	
age	0.47019	0.02518	18.675	<2e-16	***

Linear regression w/ adjustment variable (2)

- Linear regression:

- **Outcome:** Blood pressure
- **Exposure:** Treatment status
- **Adjustment covariate:** Age

$$\text{Blood pressure} = \beta_0 + \beta_1 \cdot \text{treatment} + \beta_2 \cdot \text{age}$$

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	91.86909	1.13694	80.804	<2e-16 ***
treatmentTreated	-0.58538	0.77626	-0.754	0.451
age	0.47019	0.02518	18.675	<2e-16 ***

p -value : no evidence of an association b/w treatment and BP after adjusting for age

β_0 : estimated BP for untreated, age = 0

β_1 : mean difference in BP treated vs untreated holding age constant (adjusted for age)

β_2 : mean diff in BP comparing groups differing in 1 year of age in the same treatment group

Logistic regression overview

- Linear regression – continuous outcomes (e.g. disease severity score)

- Model:

$$SeverityScore = \beta_0 + \beta_1 * TxDose$$

- Obtain estimates for coefficients β_0, β_1

- β_0 : mean outcome when $TxDose = 0$, β_1 : difference in mean outcomes comparing two groups differing in 1 unit of dose

- Logistic regression – binary outcomes (e.g. disease incidence)

- Model:

$$\log(\underline{OddsDisease}) = \beta_0 + \beta_1 * TxDose$$

- Obtain estimates for coefficients β_0, β_1
- How can we interpret β_0, β_1 ?

Logistic regression w/ binary variable (1)

- We are trying to determine if a **hypertension** is associated with **treatment status**. We obtain data from an observational study with both variables.
- We can analyze this using logistic regression:
 - **Outcome:** **Hypertension (binary)** $\leftarrow$ yes/no
 - **Exposure:** **Treatment status** $\leftarrow$ yes/no
- We do this in R and obtain the following output:

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.2329	0.1503	-14.856	<2e-16	***
tx	0.3099	0.2020	1.534	0.125	

Logistic regression w/ binary variable (2)

- Logistic regression:

- Outcome:** Hypertension (binary)
- Exposure:** Treatment status

$$\log(\text{odds}) = \beta_0 + \beta_1 \text{ treatment}$$

$$\Rightarrow \text{odds} = \exp(\beta_0 + \beta_1 \text{ treatment})$$

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.2329	0.1503	-14.856	<2e-16 ***
tx	0.3099	0.2020	1.534	0.125

$\exp(\beta_0)$: odds of hypertension when treatment = 0 is 0.11

Exponentiated coefficients:

	tx
(Intercept)	0.107221
	1.363275

$\exp(\beta_1)$: odds ratio comparing treated vs untreated participants is 1.36

p-value: H_0 : Odds ratio = 1
No evidence of a difference in odds of HTN comparing the two groups

$\Rightarrow$ Odds of HTN is 36% higher for treated people vs untreated

$$\text{Odds}_{\text{untreated}} = \exp(\beta_0 + \beta_1 \cdot 0) \\ = \exp(\beta_0)$$

$$\text{Odds Ratio} = \frac{\text{Odds}_{\text{treated}}}{\text{Odds}_{\text{untreated}}} \\ = \frac{\exp(\beta_0 + \beta_1)}{\exp(\beta_0)} = \exp(\beta_1)$$

Logistic regression w/ continuous variable (1)

- We are trying to determine if a **odds of hypertension** is associated with **age**. We obtain data from an observational study with both variables.
- We answer this question using logistic regression:
 - **Outcome:** Hypertension (binary)
 - **Exposure:** Age
- We do this in R and obtain the following output:

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-5.63319	0.59852	-9.412	< 2e-16	***
age	0.06767	0.01073	6.305	2.87e-10	***

- How can we interpret this output?

Logistic regression w/ continuous variable (2)

- Logistic regression:
 - **Outcome:** Hypertension (binary)
 - **Exposure:** Age

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-5.63319	0.59852	-9.412	< 2e-16	***
age	0.06767	0.01073	6.305	2.87e-10	***

Exponentiated coefficients:

(Intercept)	age
0.00357716	1.07001190

Logistic regression w/ adjustment variable (1)

- We are trying to determine if a **odds of hypertension** is associated with **treatment status** adjusting for **age**. We obtain data from an observational study with all variables.
- We answer this question using logistic regression:
 - **Outcome:** Hypertension (binary)
 - **Exposure:** Treatment status
 - **Adjustment covariate:** Age
- We do this in R and obtain the following output:

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-5.74480	0.60642	-9.473	< 2e-16	***
tx	0.26707	0.20679	1.292	0.197	
age	0.06714	0.01074	6.252	4.06e-10	***

Logistic regression w/ adjustment variable (2)

- Logistic regression:
 - **Outcome:** Hypertension (binary)
 - **Exposure:** Treatment status
 - **Adjustment covariate:** Age

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-5.74480	0.60642	-9.473	< 2e-16	***
tx	0.26707	0.20679	1.292	0.197	
age	0.06714	0.01074	6.252	4.06e-10	***

Exponentiated coefficients:

	tx	age
(Intercept)	0.003199387	1.069444178